DTU Compute
Department of Applied Mathematics and Computer Science

Occupancy-Aware Climate Control Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis



Occupancy-Aware Climate Control
Estimating Thermal Comfort Curves with Hidden Markov Models

Bachelor Thesis
7 December, 2026

By
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Approval

This thesis has been prepared over six months at the Section for Indoor Climate, Department of Civil Engineering, at the Technical University of Denmark, DTU, in partial fulfilment for the degree Master of Science in Engineering, MSc Eng.

It is assumed that the reader has a basic knowledge in the areas of statistics.

Mads Holme Håkonsson - s234936

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Signature

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Date

Abstract

ABSTRACT IS MISSING

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Mads Holme Håkonsson, MSc Civil Engineering, DTU
Creator of this thesis template.

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1 Introduction

Reduction in the usage of energy in buildings is a crucial aspect if one wants to go towards a more sustainable future. It has long been known that buildings are responseible for X% of the total energy consumption in the world, and that a large part of this energy is used for heating and cooling of the buildings. [MISSING REFERENCE] Within the consumption of energy in buildings, heating and cooling is a large contributor and is directly related to the thermal comfort of the occupants in the building. [MISSING REFERENCE]. Therefore, it is important to understand the bevahour of the occupants in the buildings at any giving time to be able to optimize the thermal comfort in the buildings.
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A Title

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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